

# Push-Pull 6CL6 RF Driver Stage

Class AB1 push-pull -- more drive headroom for the upgraded PA

## Why push-pull now

With the recent PA upgrade, single-ended 6CL6 was running near its plate dissipation limit. Going to a push-pull pair of 6CL6s splits the load between two tubes, giving substantial headroom while improving linearity and even-harmonic suppression in the driver itself.

## Architecture

Two 6CL6 tubes, mirror-image layout. Each tube drives one half of an interstage transformer with center-tapped primary AND center-tapped secondary. The secondary feeds the PA grids directly, with the secondary CT providing the bias return path. **This new Tr\_int replaces the existing Tr2 in the PA subcircuit.**

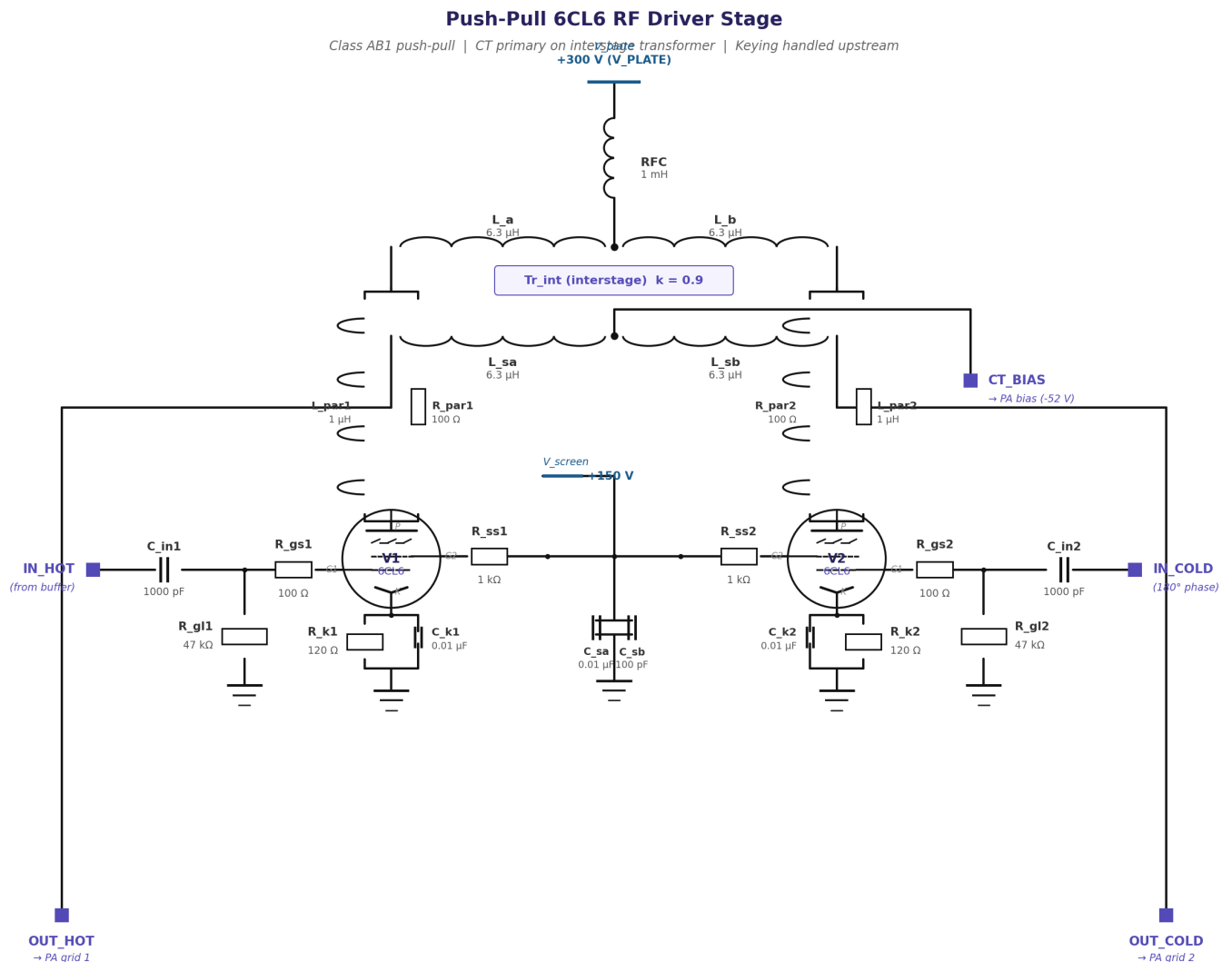


Figure 1 -- Push-pull 6CL6 driver schematic. Signal flow: IN\_HOT/IN\_COLD enter from the sides, drive the two grids through coupling caps and grid stoppers. Plates connect to the primary of Tr\_int via plate parasitic suppressors. Primary CT goes through RFC to B+. Secondary endpoints route to OUT\_HOT/OUT\_COLD ports at the bottom; secondary CT becomes the CT\_BIAS port for the PA bias return.

## Operating Point

| Parameter                  | Value (per tube)           |
|----------------------------|----------------------------|
| V <sub>p</sub> (plate)     | +300 V                     |
| V <sub>g2</sub> (screen)   | +150 V                     |
| V <sub>K</sub> (cathode)   | +3 V (self-bias)           |
| V <sub>g1</sub> (grid)     | −3 V (relative to cathode) |
| I <sub>p</sub> idle        | ≈25 mA per tube            |
| I <sub>p</sub> peak        | ≈60 mA per tube            |
| I <sub>g2</sub> idle       | ≈5 mA per tube             |
| Plate dissipation idle     | 7.5 W per tube             |
| Class                      | AB1 push-pull              |
| Combined output capability | ≈10 W to PA grids          |

## Component Values

| Ref                                   | Value                    | Notes                                         |
|---------------------------------------|--------------------------|-----------------------------------------------|
| V1, V2                                | 6CL6                     | 9-pin beam pentode pair, matched              |
| C <sub>in1</sub> , C <sub>in2</sub>   | 1000 pF / 500 V          | Grid coupling caps                            |
| R <sub>gl1</sub> , R <sub>gl2</sub>   | 47 kΩ / 0.5 W            | Grid-leak return to ground                    |
| R <sub>gs1</sub> , R <sub>gs2</sub>   | 100 Ω / 0.5 W            | Grid stoppers at G1 pin                       |
| R <sub>k1</sub> , R <sub>k2</sub>     | 120 Ω / 1 W              | Cathode self-bias resistors                   |
| C <sub>k1</sub> , C <sub>k2</sub>     | 0.01 μF / 50 V           | Cathode bypass caps                           |
| R <sub>ss1</sub> , R <sub>ss2</sub>   | 1 kΩ / 2 W               | Screen series stoppers                        |
| C <sub>sa</sub>                       | 0.01 μF / 200 V          | Screen HF bypass (shared bus)                 |
| C <sub>sb</sub>                       | 100 pF / 200 V           | Screen VHF bypass (shared bus)                |
| L <sub>par1</sub> , L <sub>par2</sub> | 1 μH                     | Plate parasitic suppressor chokes             |
| R <sub>par1</sub> , R <sub>par2</sub> | 100 Ω / 2 W              | Plate parasitic damping resistors             |
| RFC                                   | 1 mH / 100 mA            | CT-to-B+ choke                                |
| Tr <sub>int</sub>                     | CT primary, CT secondary | Interstage transformer; replaces existing Tr2 |

## Keying Strategy

---

Keying happens in the buffer/phase-splitter stage that produces IN\_HOT and IN\_COLD. The driver itself runs continuously when the transmitter is powered. When the keying stage is open, no drive reaches the driver grids, the driver tubes draw only their idle plate current, and no RF reaches the PA.

### Advantages of upstream keying:

- \* Driver simplified -- no KEY switch in cathode or grid
- \* Bias network simpler -- both tubes share a clean self-bias setup
- \* The keying shaper (RC envelope) sees small-signal levels, much easier to design than driving a high-power cathode switching transistor
- \* No transient bias shift in the driver during keying

## Layout Considerations

---

Push-pull operation depends on physical symmetry. Important items for the real build:

1. Place V1 and V2 side-by-side with mirror-image wiring
2. Match plate lead lengths to within ~1/2 inch
3. Common screen bypass should be centered between the tubes, with equal-length leads to each tube's screen pin
4. RFC and B+ feed at the primary CT should be physically centered
5. Use matched-pair 6CL6 tubes if possible. If not, add small trimmers (200  $\Omega$  in series with each R\_k) and adjust for equal plate currents

## Integration with PA Subcircuit

---

Required changes to PA\_Subcircuit.sch:

- \* **Remove the existing Tr2** (L1, L2, L3 and their three K-coupling elements)
- \* **Remove R12 and R13** (the 1 $\Omega$  series resistor and 1 M $\Omega$  DC reference that were associated with the old Tr2 primary)
- \* **Update PA ports** from {IN\_HOT, IN\_COLD, OUT\_HOT, OUT\_COLD} to {GRID\_1, GRID\_2, BIAS\_IN, OUT\_HOT, OUT\_COLD}
- \* **Wire the new GRID\_1 port directly to V1 grid** (through the existing R2 grid stopper and C3 coupling cap which stay in place)
- \* **Wire BIAS\_IN through the existing R4/R8 bias isolation resistors** to the grid bias junctions
- \* **Same for V2 side:** GRID\_2 to V2 grid, BIAS\_IN reaches both via the existing isolation resistors

### Top-level wiring in Xmitter\_top.sch:

Driver\_PP\_Subcircuit.sch ports {OUT\_HOT, OUT\_COLD, CT\_BIAS} connect to PA\_Subcircuit.sch ports {GRID\_1, GRID\_2, BIAS\_IN}. The -52 V bias supply at the top level connects to CT\_BIAS directly (the secondary CT of Tr\_int will feed it through to both PA grids via the in-PA bias network).

## Simulation Plan

---

1. Build Driver\_PP\_Subcircuit.sch with the components shown
2. Test the driver in isolation first: load OUT\_HOT/OUT\_COLD with a 10 k $\Omega$  differential resistor (representing the PA grid loading), drive IN\_HOT and IN\_COLD with two anti-phase sine sources at 0.1 V, 14.2 MHz. Verify clean differential output at

~30-50 V pk-pk

3. Adjust Tr\_int turns ratio if needed for the load impedance
4. Modify PA\_Subcircuit.sch as described, save
5. Update Xmitter\_top.sch with the new driver-PA wiring
6. Use the same IC and NODESET pattern we developed for the PA, but now also need IC values for the driver plate nodes ( $\approx +300$  V) and cathode nodes ( $\approx +3$  V)

## Tube Model Note

---

Verify tube.lib has a 6CL6 model. If not, the 12BY7A model is a close electrical equivalent and can be substituted for simulation purposes. A 6AG7 model would also work with slightly different bias point.